

European honey bee

Apis mellifera

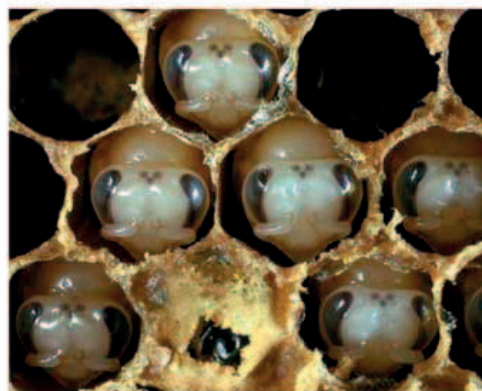
Honeybees pollinate many flowering plants, including dozens of human food plants. They are social insects that live in wild colonies and in commercial hives, which are kept for honey production. Their natural range includes Africa, Europe, and the Middle East, but they have been introduced commercially into most parts of the world. Each colony is founded by a single queen, who rears her infertile daughters as workers. The workers extend and maintain the nest, raise more sisters, and make foraging trips to collect nectar and pollen from flowers.

Division of labor

A nest is typically built in a tree hollow. It consists of honeycomb—sheets of hexagonal cells made from wax. The cells are used as nurseries for larvae (and later, pupae) and for storing nectar and pollen. Honey is produced by workers regurgitating nectar and then fanning it until it dries out. Pollen is stored separately and provides food for the developing larvae. Honey is the main source of food for the rest of the colony.

Worker bees are infertile females that live for around four to five weeks. As they age, they graduate from duties in the nest to taking off on nectar-gathering flights to flowers. Foragers communicate the location of flowers to other workers back at the colony by a figure of eight “dance.” In winter, numbers fall to 5,000, with the workers surviving on honey stores.

When a colony reaches a certain size, the queen flies off, taking half the workers with her. She leaves behind a new queen who flies off and mates with several males, collecting enough sperm to sustain her through a life that can be as long as five years, before flying back and taking over the colony.



△ WORKER BEE PUPAE

It takes 11 days for a worker bee egg to transform into a larva and then a pupa. The adult emerges at 21 days.

80,000 bees live in a colony in summer